

Using Python Libraries

Based on CBSE Curriculum

Class-12

Lesson-3

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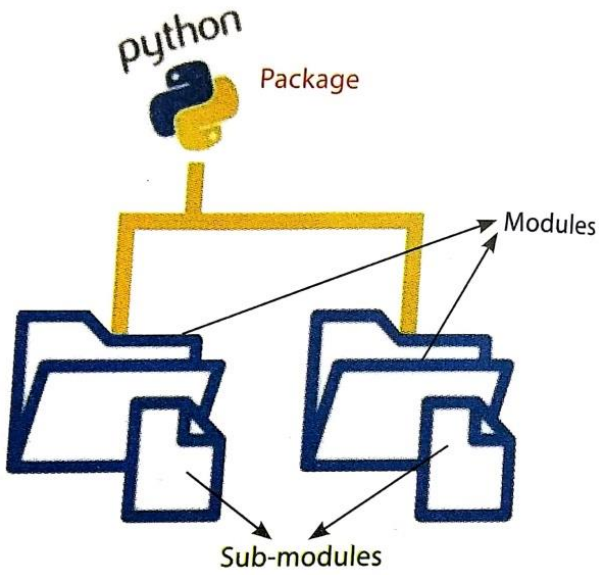
Python Libraries

- Frequently used modules are generally known as libraries which contain code for general purpose.
- These libraries are the collection of methods, classes which can be used easily.
- Python program is made using 3 different components. -
  - Library or package
  - Module
  - Functions/sub-modules



Relation Between Python Libraries, Module and Package

- A module is a file containing python definition, functions, variables, classes and statements. The extension of this file is ".py".
- While Python package, is directory(folder) of python modules.
- A library is collection of many packages in python. Generally there is no difference between python package and python library.



Module in Python

- A module is a file containing python definition, functions, variables, classes and statements. The extension of this file is ".py".
- The name of module is the name of .py file e.g. in math.py the module name is math. And ***_name_*** variable is used to store this module.
- Advantages–
  - Its biggest advantage is that we can import its functionality in any program and use it.
  - Reusability is one of the biggest advantages.
  - It helps in logically organization of Python code.
  - Programming becomes very easy when we use the collection of same types of codes.
  - Categorization : same attributes can be stored in one module.

Processing of import <module> command

- When you run the import <module> command then the following things occur -
  1. Imported module's code gets executed after interpretation.
  2. All the programs and variables of imported module are present in the program.
  3. A namespace setup is created for the imported module in the name of module.

Processing of from <module> import <object> command

- When you run the from<module> import <object> command then the following things occur -
  1. Imported module's code gets executed after interpretation.
  2. Only asked programs and variables of imported module are present in the program.
  3. No namespace is created. Import objects of module are attached to current namespace.

How to Make modules in Python?

- To make a module in python you have to make a .py file which has a name. Suppose we make a file “Shape.py”.
- Then we write different functions of area within it.
- And we put it within same folder where our program is stored.

shape.py - C:/Users/KVBBKServer/

File Edit Format Run Options Window Help

```
# This is a module named shape

def AreaCircle(r):
    return 3.14*r*r

def AreaRectangle(a,b):
    return a*b

def AreaCylinder(r,h):
    return 3.14*r*r*h
```

This is module.

*calc.py - C:/Users/KVBBKServer/

File Edit Format Run Options Window Help

```
import shape

print(shape.AreaCircle(10))
print(shape.AreaRectangle(10,20))
print(shape.AreaCylinder(10,7))
```

We used the module in our program by using import keyword. To use the members of module we use (.) as shown in the example.

```
 RESTART: C:/
314.0
200
2198.0
```

Output

How to Make modules in Python?

- Using from <module> import <function_name>

shape.py - C:/Users/KVBBKServer

File Edit Format Run Options Window Help

```
# This is a module named shape

def AreaCircle(r):
    return 3.14*r*r

def AreaRectangle(a,b):
    return a*b

def AreaCylinder(r,h):
    return 3.14*r*r*h
```

This is module.

calc.py - C:/Users/KVBBK

File Edit Format Run Options Window Help

```
from shape import AreaCircle

print(AreaCircle(10))
```

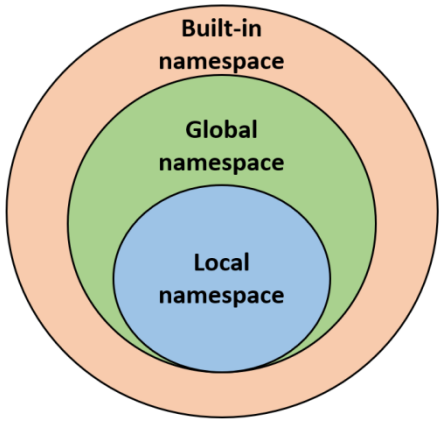
```
 RESTART:
314.0
>>>
```

Output

By using "from shape import AreaCircle "only AreaCircle function is imported and to call this there is no need to use (.) operator.

Namespaces in Python

- We imported a module in to a program which was referred as a namespace.
- To define a Namespace it is necessary to define its name.
- Python name is a kind of identifier which we use to access any python object. And in Python each and everything is an object.
- Namespaces is used to separate the different sections of a program.
- Python has 3 types of namespaces -
  - Global
  - Local
  - Built-in



Type of Namespaces

Namespaces in Python. .

- This is just like variable scope in Python.
- When we start interpreter Then a namespace is created in which print( ) and id( ) etc. are already exist. This holds all the built-in name.
- Each module creates its own global namespace.
- When we call a function then a local python namespace is created where all the names of functions are exist.

Resolving the scope of a name

- For every name reference python always follows name resolution rule when you access varaibles from any program or function.
- This is known as ***LEGB rule .*** Following works are done when this rule is followed -
  - First variable is checked in **Local Environment**, and if it is available then it will be used.
  - Then variable is checked in **Enclosing Environment**, and if it is available then it will be used.
  - Then variable is checked in **Global Environment**, and if it is available then it will be used.
  - Then variable is checked in **Built-in Environment**, and if it is available then it will be used.
  - Other wise Python will generate an error.

Module Aliasing in Python

- When you import module in Python program then we have to use the following syntax to make alia name of module -

import <ModuleName> as <AliaName>

- Then we use (.) operator to use the members of the module with alia name of module. Example is given below -

shape.py - C:/Users/KVBBKServer

File Edit Format Run Options Window Help

```
# This is a module named shape

def AreaCircle(r):
    return 3.14*r*r

def AreaRectangle(a,b):
    return a*b

def AreaCylinder(r,h):
    return 3.14*r*r*h
```

This is module.

Creating and Using the Alia Name of Module with in a program.

calc.py - C:\Users\KVBBKServer\AppData\Local\Prograr

File Edit Format Run Options Window Help

```
import shape as sp

r=int(input("Enter the radius : "))
print("Area of Circle is : ", sp.AreaCircle(r))
```

Output

```
 RESTART: C:\Users\KVBBKSer
Enter the radius : 10
Area of Circle is :  314.0
```

Module Aliasing in Python

- When you import member of any module with alia name then you have to use the following syntax -

from <ModuleName> import <MemberName> as <AliaName>

shape.py - C:/Users/KVBBKServer/

File Edit Format Run Options Window Help

```
# This is a module named shape

def AreaCircle(r):
    return 3.14*r*r

def AreaRectangle(a,b):
    return a*b

def AreaCylinder(r,h):
    return 3.14*r*r*h
```

This is module.

Creating and using of alia name of member of module.

calc.py - C:\Users\KVBBKServer\Appl

File Edit Format Run Options Window Help

```
from shape import AreaCircle as AC

r=int(input("Enter the radius : "))
print("Area of Circle is : ", AC(r))
```

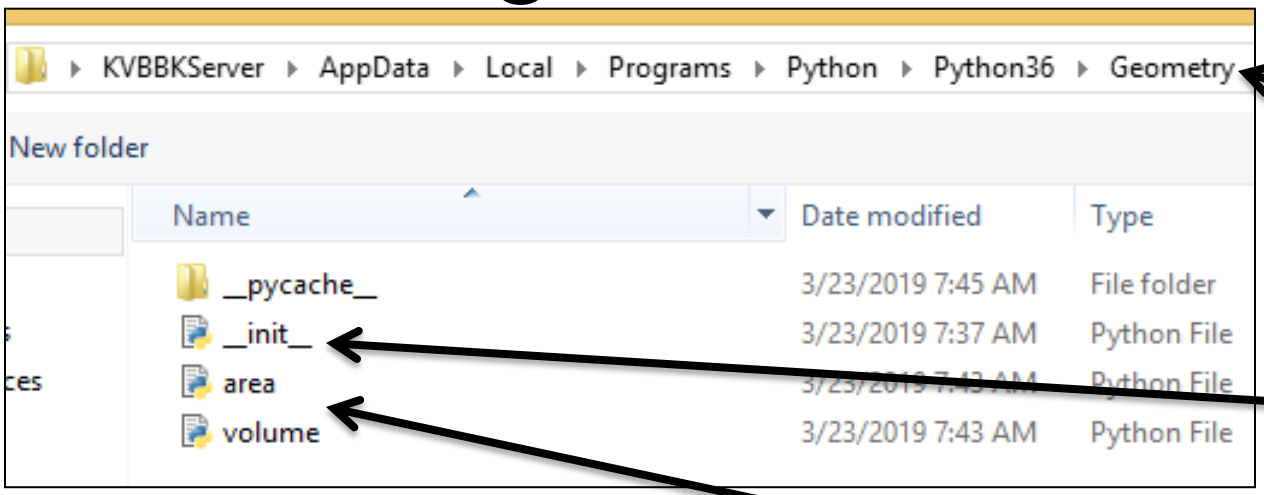
Output

```
 RESTART: C:\Users\KVBBKSer
Enter the radius : 10
Area of Circle is :  314.0
```

PACKAGE / LIBRARY

- Python Package is the collection of same type of modules.
- You can use built in package as well as your own package.
- Python Package is a simple directory. Modules are stored in that directory.
- Package or library are generally same.
- Steps to make a package is as follows – (geometry)
  1. We create a folder (directory) named geometry which will contain two files area.py and volume.py
  2. In the same directory we will put another file named **"__init__.py"**
  3. __init__.py is necessary because this is the file which tells Python that directory is package. And this file ***initializes the package***
  4. Then we import the package and use the content.

Making PACKAGE / LIBRARY

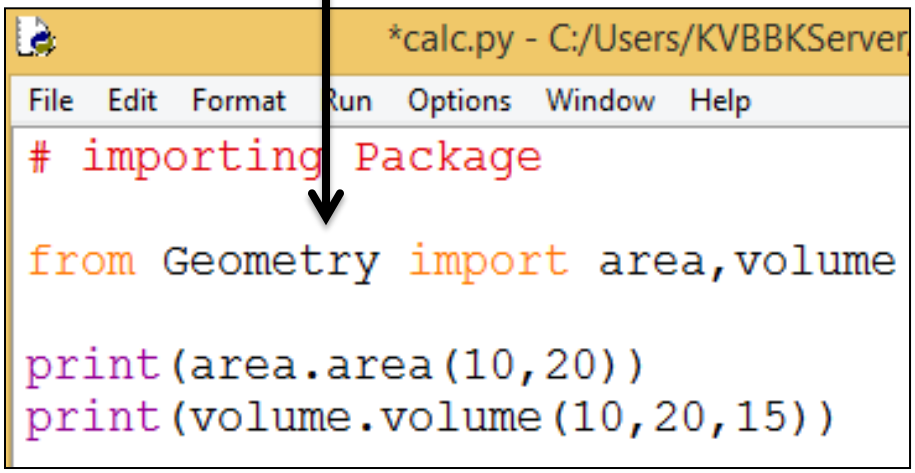


This is the folder which will be our package.

This is __init__.py file which is made directly by saving from python IDLE

These are the modules which are the members of the package.

Here the Package is used.



Output

```
 RESTART: C:/Use
200
3000
```

Locating The Module

- When we import a module then interpreter of python searches the module in the following sequence –

1. First current directory
2. If module not found then it will be searched in shell variable PYTHONPAT.
3. If not found anywhere then python searches in default. Default path is the path where Python is installed.

Standard Python Libraries

- In last chapter we discussed about math and string module. Same way we will discuss two more important libraries. - **datetime library या datetime module.**
- Python supports yyyy-mm-dd format for date. It has tow important classes -

  - date class
    - today ( )
    - year( )
    - month ( )
    - day ( )
  - time class
    - now ( )
    - hour ( )
    - minute ( )
    - second ( )

Standard Python Libraries

Date Class

```
>>> import datetime as dt
>>> dob=dt.date.today()
>>> print("Date of birth is : ",dob)
Date of birth is :  2019-03-23
>>> print("Year is : ", dob.year)
Year is :  2019
>>> print("month is : ", dob.month)
month is :  3
>>> print("Day is : ", dob.day)
Day is :  23
```

Time Class

```
>>> import datetime as dt
>>> tm=dt.datetime.now()
>>> print("Time is : ", tm)
Time is :  2019-03-23 08:09:18.673627
>>> print("Hour is : ", tm.hour)
Hour is :  8
>>> print("Minute is : ", tm.minute)
Minute is :  9
>>> print("Second is : ", tm.second)
Second is :  18
```

```
>>> import datetime as dt
>>> t=dt.time(11,30,12)
>>> print("time is : ", t)
time is :  11:30:12
```

Passing arguments in Time Class.

Thank you

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